2ND SEMESTER LAB REPORT

TITLE : Electrical Circuit Analysis Lab



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BACHELOR OF COMPUTER ENGINEERING (Fall-2025)

Submitted To:

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EXPERIMENT NO 9

Lab Title:Super Position Principle

Objectiv:Superpostion Principle

Student Name: **Muhammad Waqas -Muhammad Zeesha -Faiq Ali**

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks:

Date: 18/March/2025 Signature:

LABORATORY EXPERIMENT NO. 9

TO VERIFY SUPER POSITION PRINCIPLE

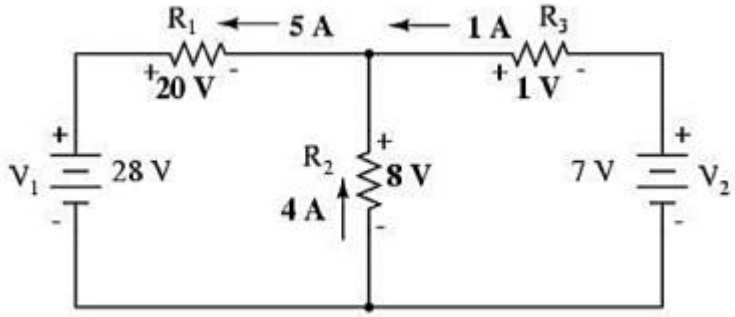
OBJECTIVE:

1. To verify Superposition Principle for DC circuits.
2. To verify using Proteus simulation.

Superposition Principle

In electric network analysis, the fundamental rules are Ohm's Law and Kirchoff's Laws. While these laws may be applied to analyze just about any circuit configuration, there are some "shortcut" methods of analysis to make the calculations easier.

"The total effect of multiple contributing sources on a linear circuit is equal to the sum of the individual effects of the sources, taken one at a time.**"**



Lab Tasks:

Lab task 1:

1. Solve using Superposition Principle:

a) Take five resistor, according to the values mentioned in the figure.

b) Remove E2 and E3 and replace them with jumpers.

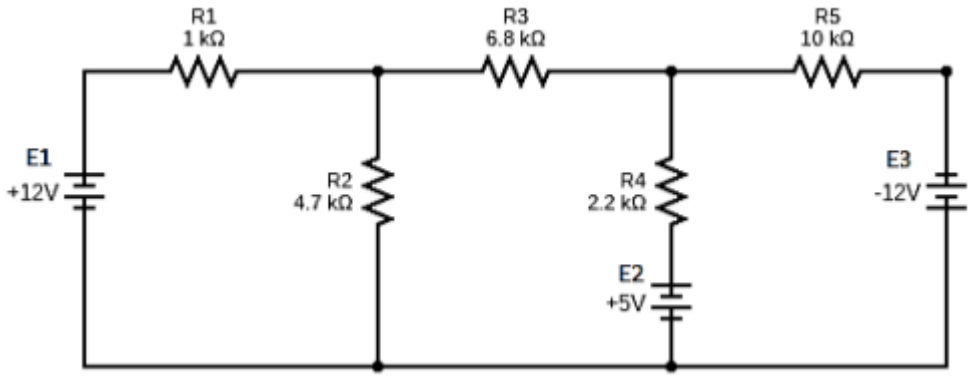
c) Measure the voltage drop across each resistor and record the values in the table.

d) Reconnect E2 and disconnect E1 and measure the voltage drops across each resistor. Record the values in the table.

e) Reconnect E3 and disconnect E2 and measure the voltage drops across each resistor. Record the values in the table.

f) Reconnect all three voltage sources and measure the voltage drops across each resistor. Record the values in the table.

g) Calculate the theoretical voltage drops using one source at a time and compare with the experimental values



Resistors	Resistance		Voltage Drop Calculated				Current Calculated			
	Rated	Measured	E1 (V)	E2 (V)	E3 (V)	E1, E2, E3	I1 (mA)	I2 (mA)	I3 (mA)	I1, I2, I3 (mA)
R1	1K Ω	0.988K Ω	2.7V	-0.37	0.19	2.80	2.91m	0.3m	0.18	2.80
R2	4.7K Ω	4.760K Ω	9.5V	0.35	-0.18	9.19	1.91m	0.3m	0.04	1.96
R3	6.8K Ω	6.77K Ω	7.4V	-2.56	1.6V	5.74	1.12m	-2.9	0.23	0.84
R4	2.2K Ω	2.2K Ω	1.7V	-1.64	-1.7V	1.54	0.83m	-1.6	0.79	-0.70
R5	10K Ω	10.1K Ω	1.8V	3.35	10V	15.4	0.18m	3.3m	1.02	1.55

Resistors	Resistance		Voltage Drop Calculated				Current Calculated			
	Rated	Measured	E1 (V)	E2 (V)	E3 (V)	E1, E2, E3	I1 (mA)	I2 (mA)	I3 (mA)	I1, I2, I3 (mA)
R1	1K Ω	0.988K Ω	2.9V	-0.35	0.18	**2.80**	2.9m	0.3m	0.18	**2.80**
R2	4.7K Ω	4.760K Ω	9.1V	0.35	-0.18	**9.19**	1.9m	0.3m	0.04	**1.96**
R3	6.8K Ω	6.77K Ω	7.1V	-2.9	1.5V	**5.74**	1.1m	-2.9	0.23	**0.84**
R4	2.2K Ω	2.2K Ω	1.8V	-1.6	-1.7V	**1.54**	0.8m	-1.6	0.79	**-0.70**
R5	10K Ω	10.1K Ω	1.8V	3.31	10V	**15.4**	0.18m	3.3m	1.02	**1.55**

E_1 = 12V **E_2** = 5V **E_3** = -12V **R_1** = 1kΩ
R_2 = 4.7kΩ **R_3** = 6.8kΩ **R_4** = 2.2kΩ **R_5** = 10kΩ

CASE 1: Only E_1 = 12V active (E_2, E_3 = 0)

Node equations: $(v_{a1} - 12)/R_1 + v_{a1}/R_2 + (v_{a1} - v_{\beta1})/R_3 = 0$ $(v_{\beta1} - v_{a1})/R_3 + v_{\beta1}/R_4 + v_{\beta1}/R_5 = 0$

Solving: v_{a1} = 9.029341V $v_{\beta1}$ = 1.892583V

Voltage drops: V_R1^(1) = 12 - v_{a1} = 2.970659V

V_R2^(1) = v_{a1} = 9.029341V

V_R3^(1) = v_{a1} - $v_{\beta1}$ = 7.136758V

V_R4^(1) = $v_{\beta1}$ = 1.892583V

V_R5^(1) = $v_{\beta1}$ = 1.892583V

Currents:

I_R1^(1) = 2.970659/1k = 0.002971A = 2.971mA

I_R2^(1) = 9.029341/4.7k = 0.001921A = 1.921mA

I_R3^(1) = 7.136758/6.8k = 0.001050A = 1.050mA

I_R4^(1) = 1.892583/2.2k = 0.000860A = 0.860mA

I_R5^(1) = 1.892583/10k = 0.000189A = 0.189mA

CASE 2: Only E_2 = 5V active (E_1, E_3 = 0)

Node equations:

$v_{a2}/R_1 + v_{a2}/R_2 + (v_{a2} - v_{\beta2})/R_3 = 0$ $(v_{\beta2} - v_{a2})/R_3 + (v_{\beta2} - 5)/R_4 + v_{\beta2}/R_5 = 0$

Solving:

v_{a2} = 0.358444V

$v_{\beta2}$ = 3.314460V

Voltage drops:

V_R1^(2) = -0.358444V

V_R2^(2) = 0.358444V

V**_R3^(2)** = -2.956017V

V_R4^(2) = -1.685540V

V_R5^(2) = 3.314460V

Currents:

I_R1^(2) = -0.358444/1k = -0.000358A = -0.358mA

I**_R2^(2)** = 0.358444/4.7k = 0.000076A = 0.076mA

I_R3^(2) = -2.956017/6.8k = -0.000435A = -0.435mA

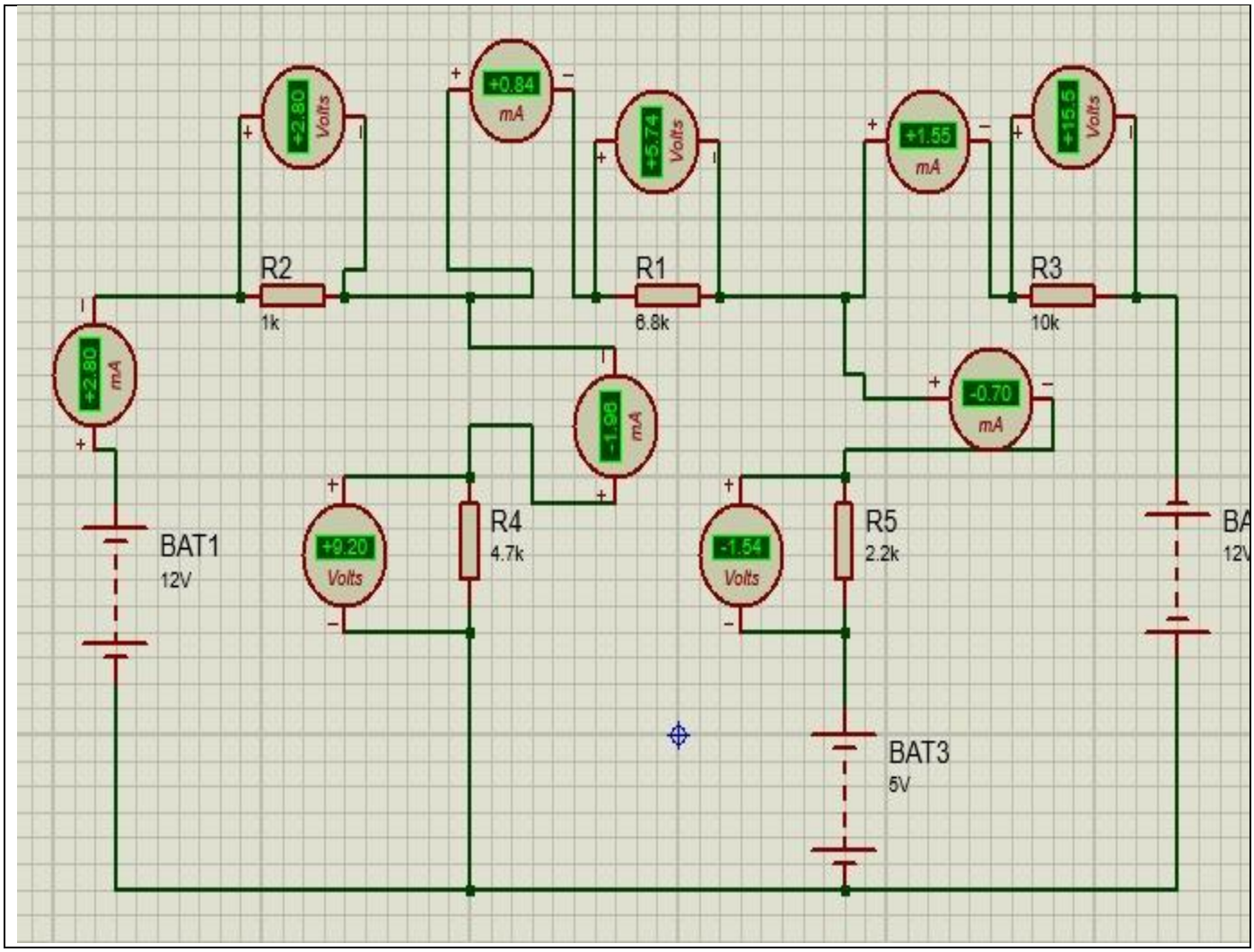
I_R4^(2) = -1.685540/2.2k = -0.000766A = -0.766mA

I_R5^(2) = 3.314460/10k = 0.000331A = 0.331m

CASE 3: Only E_3 = -12V active (E_1, E_2 = 0) Node equations: $v_{a3}/R_1 + v_{a3}/R_2 + (v_{a3} - v_{\beta3})/R_3 = 0$ $(v_{\beta3} - v_{a3})/R_3 + v_{\beta3}/R_4 + (v_{\beta3} + 12)/R_5 = 0$ Solving: v_{a3} = -0.189258V $v_{\beta3}$ = -1.750035V Voltage drops: **V_R1^(3)** = 0.189258V **V_R2^(3)** = -0.189258V **V_R3^(3)** = 1.560777V **V_R4^(3)** = -1.750035V **V_R5^(3)** = 10.249966V Currents: **I_R1^(3)** = 0.189258/1k = 0.000189A = 0.189mA **I_R2^(3)** = -0.189258/4.7k = -0.000040A = -0.040mA **I_R3^(3)** = 1.560777/6.8k = 0.000229A = 0.229mA **I_R4^(3)** = -1.750035/2.2k = -0.000796A = -0.796mA **I_R5^(3)** = 10.249966/10k = 0.001025A = 1.025mA	**FINAL TOTAL VALUES (Superposition):** Voltages: **V_R1** = 2.970659 + (-0.358444) + 0.189258 = 2.801474V **V_R2** = 9.029341 + 0.358444 + (-0.189258) = 9.198526V **V_R3** = 7.136758 + (-2.956017) + 1.560777 = 5.741518V **V_R4** = 1.892583 + (-1.685540) + (-1.750035) = -1.542992V **V_R5** = 1.892583 + 3.314460 + 10.249966 = 15.457008V **Currents:** **I_R1** = 2.801474/1k = 0.002801A = 2.80mA **I_R2** = 9.198526/4.7k = 0.001957A = 1.96mA **I_R3 =** 5.741518/6.8k = 0.000844A = 0.84mA **I_R4 =** -1.542992/2.2k = -0.000701A = -0.70mA **I_R5** = 15.457008/10k = 0.001546A = 1.55mA

As values give in table after calculation values are exactly same as give in protius simulation

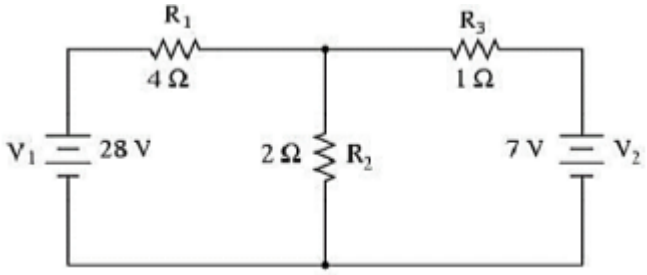
Proteus Simulation:



Lab Task 2:

Solve the following circuit by Superposition Principle:

For the circuit explained and solved in the manual earlier, do the calculation yourselves and simulate it on Proteus. Also, perform the task on hardware and verify your results.



Resistors	Resistance		Voltage Drop Measured			Current Measured		
	Rated	**Measured**	**V1**	**V2**	**V1,V2**	**I1**	**I2**	**I1, I2**
R1	4 ohm	4 ohm	24 V	-4V	**20V**	6A	-1A	**5A**
R2	2 ohm	2.02 ohm	4V	4V	**8V**	2A	2A	**4A**
R3	1 ohm	1.0 ohm	4V	-3V	**1V**	4A	-3A	**1A**

Resistors	Resistance		Voltage Drop Measured			Current Measured		
	Rated	**Measured**	**V1**	**V2**	**V1,V2**	**I1**	**I2**	**I1, I2**
R1	4 ohm	4 ohm	24 V	-4V	**20V**	6A	-1A	**5A**
R2	2 ohm	2.02 ohm	4V	4V	**8V**	2A	2A	**4A**
R3	1 ohm	1.0 ohm	4V	-3V	**1V**	4A	-3A	**1A**

Theoretical Calculation:

Superposition Principle Analysis

Step 1: Only V1 (28V) Active

- Equivalent resistance parallel to V1: Req = R2 || (R3) = 2 || 1 = 2×1/(2+1) = 2/3 Ω
- Total resistance: Rtot = R1 + Req = 4 + 2/3 = 14/3 Ω
- Total current: I = V1/Rtot = 28/(14/3) = 6A
- Voltage across R2||R3: V = I×Req = 6×(2/3) = 4V
- Current through R2: IR2(1) = V/R2 = 4/2 = 2A
- Current through R3: IR3(1) = V/R3 = 4/1 = 4A

Step 2: Only V2 (7V) Active

- Equivalent resistance parallel to V2: Req = R1 || R2 = 4 || 2 = 4×2/(4+2) = 4/3 Ω
- Total resistance: Rtot = Req + R3 = 4/3 + 1 = 7/3 Ω
- Total current: I = V2/Rtot = 7/(7/3) = 3A
- Voltage across R1||R2: V = I×Req = 3×(4/3) = 4V
- Current through R1: IR1(2) = V/R1 = 4/4 = 1A (opposite direction)
- Current through R2: IR2(2) = V/R2 = 4/2 = 2A

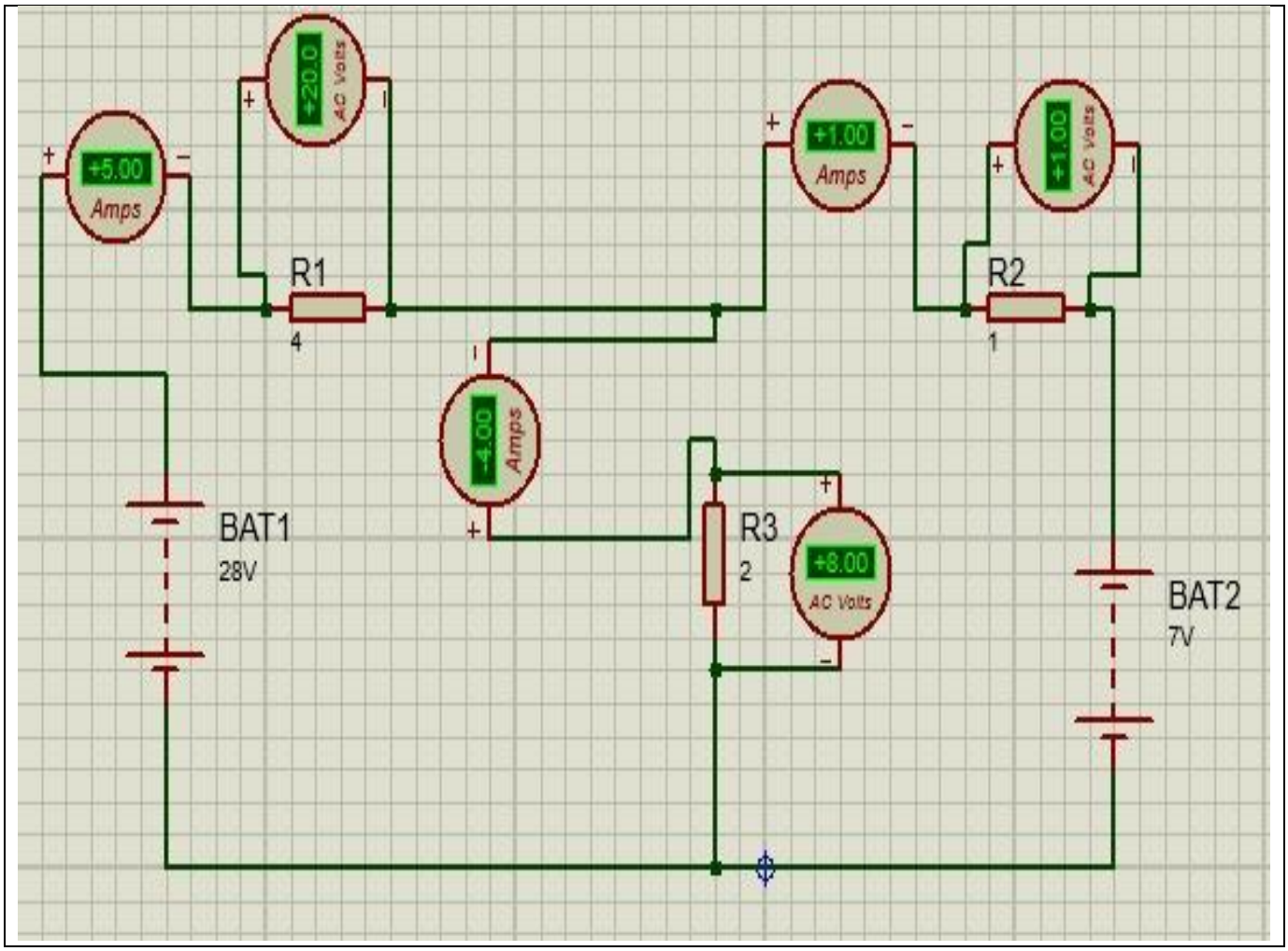
Step 3: Combine Results

- **IR1 = 6A - 1A = 5A**
- **IR2 = 2A + 2A = 4A**
- **IR3 = 4A - 3A = 1A**

Step 4: Calculate Voltages

- **VR1 = IR1×R1 = 5A×4Ω = 20V**
- **VR2 = IR2×R2 = 4A×2Ω = 8V**
- **VR3 = IR3×R3 = 1A×1Ω = 1V**

Proteus Simulation:



Conclusion:

The Superposition Principle helps us understand circuits with more than one power source. It says that the total voltage or current in any part of a circuit can be found by looking at one power source at a time. To do this, we keep one source active and replace all other voltage sources with wires (short circuits) and all current sources with gaps (open circuits). After analyzing the effect of each source separately, we add up all the results to get the final answer. This method works only for linear circuits and makes solving complex circuits easier and more organized.